

Hospital Management System(HMS)

Project Proposal

1. Introduction

- The Hospital Management System (HMS) is designed to modernize hospital operations through a structured database system.
- Its purpose is to reduce manual errors, streamline workflows, and facilitate better decision-making.
- Key operations managed by the system include:
 - ❖ Patient management
 - ❖ Doctor information management
 - ❖ Appointment scheduling
 - ❖ Room allocation
 - ❖ Admissions
 - ❖ Prescriptions

2. Objectives

- ❖ Design a relational database with proper normalization.
- ❖ Maintain accurate records of patients, doctors, appointments, and admissions.
- ❖ Automate hospital operations to reduce paperwork.
- ❖ Ensure data consistency using primary and foreign keys.
- ❖ Provide fast and efficient data retrieval for hospital staff.

3. Scope of the Project

The HMS project will cover:

- Patient information management
- Doctor and department management
- Appointment scheduling

- Room and admission tracking
- Prescription management and retrieval

4. System Description (Modules)

1. **Departments Module:** Stores department names and relevant details.
2. **Doctors Module:** Manages doctor profiles and specializations.
3. **Patients Module:** Stores patient demographics and medical histories.
4. **Appointments Module:** Schedules and tracks doctor–patient appointments.
5. **Rooms Module:** Monitors room types, availability, and assignments.
6. **Admissions Module:** Handles patient admissions, room allocation, and discharge processes.
7. **Prescriptions Module:** Maintains doctor-prescribed medications and dosage instructions.

5. Database Design

- ❖ The database consists of seven main tables: Departments, Doctors, Patients, Appointments, Rooms, Admissions, and Prescriptions.
- ❖ These tables are linked through primary and foreign keys to maintain relational integrity and support efficient data retrieval.
- ❖ The design follows normalization principles to reduce redundancy and ensure consistency.

6. Tools & Technologies

- **Database:** MySQL
- **Query Language:** SQL for data management
- **Storage Engine:** InnoDB
- **Testing Tools:** phpMyAdmin / OneCompiler
- **Design Tools:** ER Diagram software (optional)

7. Benefits of the System

- ✚ Minimizes manual errors and data inconsistencies.
- ✚ Automates appointment and admission tracking.
- ✚ Improves hospital operational efficiency.
- ✚ Centralizes and secures all hospital data.
- ✚ Enables quick access to patient and doctor information.

8. Expected Outcome

- ✓ A fully functional, normalized database supporting all hospital operations, including appointments, admissions, and prescriptions.
- ✓ The system can be extended in the future to include billing, reporting, and laboratory modules.

9. Conclusion

- ✚ HMS provides a robust and scalable solution for managing hospital operations efficiently.
- ✚ The project demonstrates strong SQL and relational database skills and logical organization of medical data.
- ✚ By automating workflows, it enhances operational efficiency and data, making it accuracy, suitable for real-world healthcare applications.